Chapter 33
Swaps Revisited

Chapter 7 priced the plain vanilla swap and built the curves it needs. It also deferred almost everything else, because the non-standard swaps require the adjustments of Chapter 30 and the measure machinery of Chapter 28, and those had not been developed.

They have now, and this chapter prices the deferred instruments. The organizing observation is that a non-standard swap that needs more than discounting is usually a standard swap plus a convexity adjustment, and that the adjustment is computable, has a sign that can be reasoned about, and is often large enough to matter.

There is a second reason to return to swaps here. The transition to risk-free rates changed some of these instruments profoundly and others not at all. Section 33.4 contains the sharpest case: a convexity adjustment the market spent twenty years computing became roughly sixty times smaller in the currencies that retired their IBOR, not because anyone solved it but because the underlying convention changed.

The chapter runs from the variations that need no new theory to the ones that need all of it. Sections 33.1 to 33.3 change the notional or the payment dates and price by discounting alone. Sections 33.4 to 33.6 are where convexity enters: in-arrears swaps before and after the transition, CMS swaps, and range accruals, which are a strip of digitals in disguise. Sections 33.7 to 33.10 are the swaps whose payoff references something else: an embedded option, a commodity, a basis, an inflation index. Sections 33.11 to 33.13 are the equity and total return swaps whose funding leg Chapter 27 made expensive, how a position is unwound, and who uses these.

33.1 AMORTIZING, ACCRETING, AND ROLLER-COASTER SWAPS

The simplest variation is a notional that changes over time. An amortizing swap steps the notional down, typically to match a loan being repaid; an accreting swap steps it up, matching a construction drawdown; a **roller-coaster swap** does both. Pricing needs nothing new. The schedule is known in advance, so each leg is discounted as it stands.

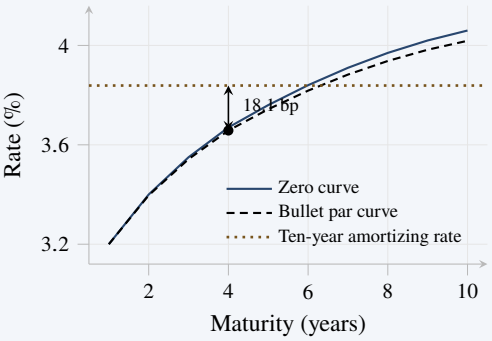
Example 33.1 Why the average-life approximation fails

Take a ten-year annual swap amortizing linearly from 100 to zero (notional 100 in the first year, 90 in the second, down to 10 in the tenth). The zero curve is upward-sloping and flattening, at 3.20, 3.40, 3.55, 3.67, 3.76, 3.84, 3.91, 3.97, 4.02 and 4.06% for one through ten years, annually compounded. The par rate is

$$\frac{\sum_k N_k \left[P(0,t_k) - P(0,t_{k+1})\right]}{\sum_k N_k \, P(0,t_{k+1})} = \mathbf{3.8386}\%$$

For comparison, here are the rates on **bullet swaps**, ordinary swaps whose notional stays at 100 until maturity:

Bullet 10-year par rate	4.0187%
Bullet 5-year par rate	3.7432%
Bullet 4-year par rate	3.6576%
10-year amortizing par rate	**3.8386%**



The tempting shortcut is to price the amortizing swap as a bullet swap at its weighted average life. Weighting each payment date by the notional outstanding over the period ending on it, that life is 4.0 years here, and the bullet par rate at four years is 3.6576%, an error of −18.1 **basis points**.

Which definition is used matters. Weight instead by principal repaid, which is what *weighted average life* means in the bond market, and the answer is 5.5 years, where the bullet rate is 3.7775% and the error is −6.1 basis points. Two standard summaries of one schedule differ by eighteen months, so the shortcut's error is not even a well-defined number until the convention is fixed.

Neither life is the right summary. The swap is a portfolio of ten bullet swaps of 10 notional each, one maturing at each repayment date, so its par rate averages the bullet par rates weighted by each one's annuity, and an annuity grows with maturity: on those weights the mean maturity is 6.9 years, and the six- and seven-year bullet rates bracket 3.8386%. Weighted average life puts the mean too early, so on a rising curve the error is first order in the slope. A straight curve does not remove it: on a zero curve rising in a straight line from 3.20% to 4.10%, the four-year shortcut misses by −26.3 basis points, and only a flat curve makes it exact.

Curvature accounts only for what is left at the right weights, and that residual is the error of Section 20.9's naive expectation: replacing a distribution by its mean inside a nonlinear function.

33.2 FORWARD-STARTING AND DELAYED SWAPS

A forward-starting swap (also called a delayed or deferred-start swap) begins at a future date. It requires no new machinery: the forward swap rate is read off the curve as in Section 7.3, and the value is zero at the forward par rate by construction.

What it requires is care about which curve. A ten-year swap starting in five years is priced from the five- and fifteen-year points, and an error of one basis point in the fifteen-year rate moves its rate by about *one and a half*, because the forward rate is a difference of two long rates divided by a shorter span, and differences amplify errors. This is why forward-starting swaps are the instrument that exposes curve-construction disagreements between desks, and why the interpolation choices of Section 7.6, which look like housekeeping, are argued about.

33.3 COMPOUNDING SWAPS

In a **compounding swap**, the floating payments are not paid as they accrue; they are reinvested at a specified rate and paid as a single amount at maturity. The reinvestment rate is the whole of the pricing question. The conventions differ in that rate, and they price differently.

Without a spread, the payments are reinvested at the floating rate itself. The leg's value is then the same as an ordinary floating leg's: the payments are rolled up at the rate they would have earned if received and reinvested, so no adjustment arises, and compounding swaps are usually treated as an operational rather than a valuation matter. The users are borrowers whose debt capitalizes interest, a construction loan or a payment-in-kind facility, so the hedge must roll up as the loan does.

Once the leg carries a spread, ISDA offers two conventions. *Flat compounding* compounds

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the accrued amounts at the floating rate alone and so keeps that property. *Compounding* in ISDA's narrower sense compounds them at the floating rate plus the spread, and then the leg's value is *not* what it would be if each amount were paid as it accrued, because the spread is being compounded at a rate that would not have been earned. The correction is small but real, and it is the reason ISDA's compounding definitions run to several pages.

Compounding at a fixed rate makes the leg a portfolio of forward rate agreements with deterministic weights: straightforward, and rare.

The general point is the one Chapter 28 supplies. A payment moved from its natural date to another date is a change of measure, and no adjustment arises only if what is paid on the new date, measured in that date's zero-coupon bond, is still a martingale. Rolling a payment forward at the floating rate itself is self-financing, so flat compounding (and compounding with no spread) passes that test; compounding with the spread and at a fixed rate do not.

33.4 IN-ARREARS SWAPS, BEFORE AND AFTER

This is the instrument the transition changed most. The classic **LIBOR-in-arrears swap** pays the rate for $[T_k, T_{k+1}]$ at T_k, at the *start* of the period it applies to, rather than the end. Since LIBOR was set in advance, this was operationally possible and was done. It was a view on the curve's shape, taken mostly when the curve was steep: the fixed rate against in-arrears LIBOR is priced off forwards one period further out, so a corporate borrower or investor receiving fixed earned a higher rate than on a standard swap, and gained if LIBOR set below the forwards.

But the rate is a martingale under the T_{k+1}-forward measure, and the payment is under the T_k one. Section 30.4 derived the resulting adjustment; to first order in $\sigma^2 T_k$ it is the following, and Example 33.2 puts numbers on it:

$$E^{T_k}[F_k] \approx F_k + \frac{F_k^2\,\sigma^2\,\tau\,T_k}{1 + F_k\tau} \tag{33.1}$$

where F_k is the forward rate for the period, τ its length in years, and σ the rate's volatility.

Example 33.2 A convexity that the transition mostly removed

Take $F = 3.5\%$, $\tau = 0.5$ and $\sigma = 25\%$:

T_k	Adjustment (first order)	Adjusted rate
0.5y	0.19 bp	3.5019%
1y	0.38 bp	3.5038%
2y	0.75 bp	3.5075%
5y	1.88 bp	3.5188%
10y	3.76 bp	3.5376%

The transition removed the mismatch rather than the volatility. A compounded-in-arrears SOFR rate is determined at the end of its accrual period and paid then, so the payment is no longer early and (33.1) has nothing to correct. What is left is the short delay between the end of the observation period and the payment date, typically two business days. That makes the payment late rather than early, so the adjustment is negative:

T_k	Adjustment at a 2-day lag
1y	−0.0061 bp
5y	−0.0304 bp
10y	−0.0607 bp

At ten years the adjustment falls from 3.76 basis points to 0.06, about sixty times smaller. That changes how the product should be classified. The rate replacing LIBOR is the overnight

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rate compounded in arrears, which is what the floating leg of Section 7.1's overnight indexed swap pays, so the instrument this section files as non-standard is now the standard one. Paying in arrears is the convention, and a rate fixed in advance is now what needs a special name and a separate construction: Chapter 4's Term SOFR. The convention traded places with its own exception.

The old product is not extinct, only currency-specific. Paying a period's rate at that period's start requires knowing it at the start, so the structure needs a forward-looking term rate, and one survives wherever the transition left it standing. EURIBOR was never retired, so a EURIBOR-in-arrears swap carries the classic adjustment unchanged: 3.76 basis points at ten years, not 0.06. Term SOFR would serve in dollars, but the ARRC's recommendation limits it to business loans and the derivatives hedging them, not the swap market at large.

A convexity adjustment is a property of a convention rather than of a market: nothing about interest rate volatility changed, the timing convention did, and the adjustment followed it—into the currencies that kept a term rate.

The new convention has a problem of its own, in a different place. Section 29.4's backward-looking caplet needs an adjustment the forward-looking version did not, because the rate is unknown until the period ends. The complexity moved rather than vanished, which is true of most convention changes in this book.

33.5 CMS SWAPS

A constant-maturity swap exchanges a fixed rate for a long swap rate (the ten-year rate, say) observed and paid periodically. It is the non-standard swap where the adjustments of Chapter 30 are unavoidable.

Example 33.3 What ignoring the adjustment costs

Pay the ten-year swap rate, observed on each payment date, annually for ten years against a fixed rate. Forward ten-year rate flat at 3.40%, swap-rate volatility 25%, discounting at 3.2%.

Section 30.5's adjustment, with G the ten-year bond including its principal as that section defines it, applies to each payment, growing linearly in its date:

Payment	Adjustment	Adjusted rate
1	3.65 bp	3.4365%
3	10.95 bp	3.5095%
5	18.25 bp	3.5825%
7	25.55 bp	3.6555%
10	36.51 bp	3.7651%

The fair fixed rate is the annuity-weighted average of the adjusted rates:

Fair fixed rate, with adjustments	3.5913%
Fair fixed rate, ignoring them	3.4000%
Difference	**19.13 bp**

The difference is nineteen basis points on the fixed leg, every year for ten years. A ten-year swap trades at a bid–ask of well under a basis point, so this is not a refinement—a desk quoting a CMS swap without the adjustment is not slightly wrong, it is not in the market.

The adjustment also depends on a volatility, which is the structural point. An ordinary swap is priced off the curve alone; a CMS swap requires a swaption volatility, so a linear-looking product carries a vega. That has three consequences.

The CMS desk is a volatility desk. Its hedges are swaptions, not swaps, and its risk report has a vega column.

The adjustment uses the whole smile, not one volatility. The replication of Section 30.5 integrates over strikes, weighting each by the payoff's second derivative (the construction Section 22.3 uses,

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with $1/K^2$ weights, to replicate variance), so the number depends on the smile's wings, where few strikes are quoted and the value comes from the SABR fit of Section 29.9 rather than from the market.

And the sign is always the same. The adjustment is positive, so a payer of CMS always pays more than the naive forward. An investor buying a CMS-linked note is being sold that convexity, and the note's headline coupon does not reveal whether the price is fair.

33.6 RANGE ACCRUALS

A **range accrual** coupon accrues only on days when a reference rate sits inside a band. It is the most widely sold rate-linked structured coupon after CMS, and it is a strip of digital options, a pair for each observation day; Example 33.4 prices a quarter's coupon.

Example 33.4 The enhancement is the short vega

A coupon accruing only on days when the three-month rate is inside $[2.50\%, 4.50\%]$, over the quarter from year 2 to year 2.25, observed daily: 90 observations at $T_i = 2 + i/360$. Forward rate 3.50%, flat across the quarter; volatility 25%.

Each observation day is a pair of digitals. Under the forward measure the rate is lognormal, so each digital is a Black-style probability. For a strike K and an observation at T_i,

$$P(R_{T_i} < K) = N(-d_2), \qquad d_2 = \frac{\ln(F/K) - \sigma^2 T_i/2}{\sigma\sqrt{T_i}}$$

and the expected accrual fraction is the average of the 90 in-range probabilities.

$P(\text{rate} < 2.50\%)$, averaged	0.2294
$P(\text{rate} < 4.50\%)$, averaged	0.8084
P(**inside**)**—the expected accrual fraction**	**0.5790**

The days are not interchangeable. The distribution keeps widening, so the in-range probability falls from 0.5931 on the first day to 0.5655 on the last, and pricing the whole quarter off a single observation at two years would give 0.5934 and understate the enhancement by about 15 basis points.

So a coupon of $3.50\%/0.5790 = 6.0447\%$, paid only on in-range days, has the same value as a vanilla 3.50% coupon. The headline enhancement is **254 basis points, 73% above the vanilla coupon**, which is what sells the note.

Now see what the investor gave up:

Realized volatility	Accrual fraction	Value of the 6.0447% coupon
15%	0.8194	4.9532%
20%	0.6865	4.1500%
25%	0.5790	**3.5000%**
35%	0.4297	2.5976%
50%	0.2996	1.8113%

At the pricing volatility the coupon is worth exactly the vanilla 3.50%, by construction. Above it the investor receives less, below it more.

The enhancement is the short vega, priced. A range accrual investor is selling volatility, and the 254 basis points is the premium for that sale, spread over the coupon rather than paid up front. Nothing is being given away.

Two structural points follow. The dealer is long volatility and hedges with a strip of digitals or, since digitals are hedged as tight call spreads, with a great many options near the barriers. Section 12.8's argument for building an arbitrary payoff is the machinery, and the practical consequence is that the dealer's hedge is concentrated exactly at the range boundaries, which is where the note's value is most sensitive and where the smile is least reliable.

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The investor's loss is also asymmetric in a way the table understates. Rate volatility rises in the dislocations (a policy shock, an inflation surprise) that also squeeze the holder's other income, so the coupon fails in the states where its holder most needs income, which is Section 27.9's wrong-way structure arriving in a retail product.

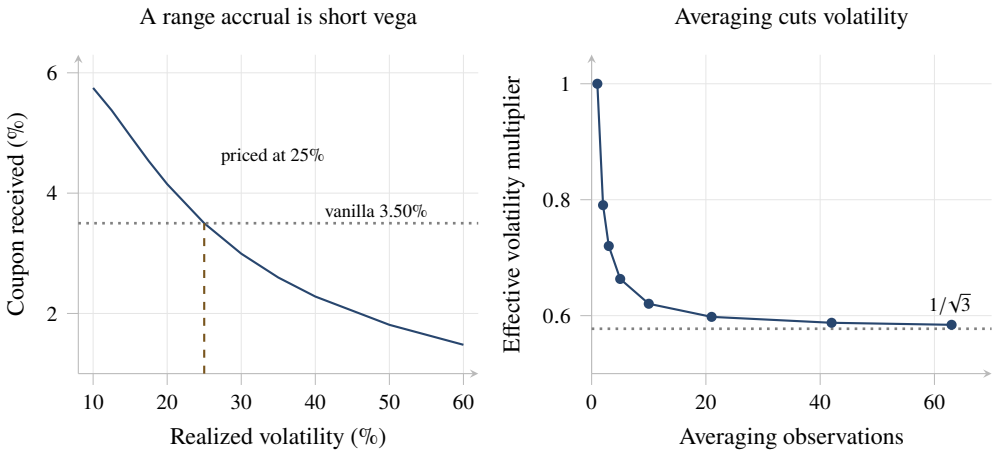


Figure 33.1: Left: Example 33.4's coupon against realized volatility. Right: the multiplier $\sqrt{(n+1)(2n+1)/6n^2}$ of Example 33.6.

Figure 33.1's left panel is the picture a range accrual prospectus does not contain. The coupon is a decreasing function of a quantity the investor is usually not forecasting when buying the note, and it falls fastest at low volatility: about 0.17 percentage points of coupon per volatility point at 15%, against 0.11 at the 25% pricing level.

33.7 CANCELLABLE AND EXTENDABLE SWAPS

A **cancellable swap** may be terminated early by one party; an **extendable swap** may be prolonged. Both are a vanilla swap plus a Bermudan swaption, and Chapter 31 priced exactly that option. Example 33.5 carries its value onto the swap by dividing it by the swap's annuity, which turns a one-off option premium into a running spread on the fixed leg.

Example 33.5 The Bermudan, annuitized

Take Example 31.8's six-year swap with annual cancellation rights from year one, whose Bermudan value was **1.6995** per 100 of notional.

Annuity of the six-year swap	5.272406
Bermudan option value	1.6995
Spread on the fixed leg	**32.23 bp**
Vanilla six-year par rate	3.9862%
Cancellable swap fixed rate	**4.3085%**

This is only the first step of an iteration. The fixed rate is the par rate plus the option's value over the annuity, but the option the borrower buys is a receiver struck at that fixed rate, so the rate the calculation produces is also one of its inputs. The right rate is the fixed point $c^* = 3.9862\% + V(c^*)/5.272406$, where $V(c)$ is the Bermudan struck at c. The table takes one step from Example 31.8's 4% strike. A second strikes the receiver at 4.3085%, deeper in the money and worth 2.3295 on Chapter 31's tree; iterating until the strike stops moving gives 4.5339%,

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a spread of 54.8 basis points, so the first iteration understates the cost by about two fifths. The iteration holds Chapter 31's a and σ fixed. A desk calibrating to co-terminal swaptions struck at the deal's own rate must recalibrate at every step, because the volatility quoted at 4.53% is not the one quoted at 4%; only the annuity and the par rate, which come from the curve, stay put.

A borrower whose debt may be prepaid or refinanced is the natural buyer, and it pays 32 basis points a year on the first iteration, and nearer 55 at the fixed point, for the right to walk away—the whole economics of the instrument. An extendable swap serves the borrower that expects to roll its debt and wants the option to keep the hedge.

The reason to put this here rather than in Chapter 31 is what it does to the calibration problem. Example 31.9 showed the Bermudan's value spanning 13.1% across mean-reversion assumptions that all reproduce the calibrating European exactly. Annuitized, that becomes

$$0.131 \times 32.23 = \mathbf{4.2} \text{ basis points}$$

of ambiguity in a quoted swap rate, on the first iteration.

Four basis points is well outside the bid–ask on a six-year swap, which is a fraction of a basis point. So the model ambiguity of Chapter 31 is not a theoretical concern that gets absorbed in spreads—it is several times the spread, and two dealers quoting the same cancellable swap from the same European volatilities can differ by more than the market they are quoting into.

33.8 COMMODITY SWAPS AND THE COST OF AVERAGING

A commodity swap exchanges a fixed price for a floating one, and the floating one is almost always an average of daily prices over the delivery month rather than a single fixing, the one published price observed on a stated date. That convention exists because the underlying physical exposure is itself an average (a refiner buys crude every day), and it has a large valuation consequence, which Example 33.6 measures.

Example 33.6 What averaging does to volatility

The average of a lognormal price over n equally spaced dates $T/n, 2T/n, \ldots, T$ has, in logs, variance

$$\frac{(n+1)(2n+1)}{6n^2}\,\sigma^2 T$$

which tends to $\sigma^2 T/3$ as n grows. This is exact for the geometric average and a close approximation for the arithmetic average a contract actually settles on. The effective volatility multiplier is therefore

n	Variance ratio	Vol multiplier
1	1.0000	1.0000
5	0.4400	0.6633
10	0.3850	0.6205
21	0.3575	**0.5979**
63	0.3413	0.5842
∞	0.3333	$1/\sqrt{3} = 0.5774$

Twenty-one fixings, at one every two and a half weeks over a year, already take the multiplier to within four percent of its continuous limit, so a modest number of fixings does essentially all the work continuous averaging would. What the multiplier does assume is that the averaging spans the option's whole life; an average over only the last month of a longer option cuts far less.

The pricing consequence is large. Consider an at-the-money one-year option on an \$80 commodity with a 35% volatility, settled either on one fixing or on 21 fixings spread across the year:

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	Effective volatility	Premium
On a single fixing at expiration	35.00%	11.1136
On the 21-fixing average	20.93%	6.6669

The averaging alone removes 40% of the option premium, with no change to the underlying's volatility. Two implications follow. First, a commodity option quoted on an average is not comparable with one quoted on a fixing, and comparing their implied volatilities without adjusting is a common and expensive error.

The second is a trap: the swap itself is not affected, while the option is. A commodity *swap* settles against the average, but its fair fixed price is the average of the forwards, a linear function, so averaging changes nothing. It is only when optionality is attached, as in a commodity cap or a swaption on the average, that the multiplier bites.

33.9 BASIS SWAPS, AND WHAT THEY PRICE

Section 7.7 introduced tenor basis swaps (three-month against six-month) and Section 7.8 the cross-currency basis. Here is what a basis swap is, in this Part's language.

A tenor basis swap prices the difference between two forecast curves. Before 2022 that difference was largely credit and liquidity: three-month LIBOR embedded less bank credit risk per unit time than six-month. After the transition to overnight rates compounded over the relevant period, the credit component is gone in the currencies that retired their IBOR, and what remains there is a much smaller basis reflecting term premia and the mechanics of compounding. EURIBOR kept the credit component: three-month against six-month EURIBOR, and EURIBOR against €STR, still price bank credit. The tenor basis market therefore shrank rather than transformed, a rare case of a market being simplified.

The cross-currency basis did not shrink. It prices the relative scarcity of funding in two currencies, which is not a convention and does not go away. Section 7.8's account stands, and Section 9.6's ColVA is its valuation-side counterpart: a CSA permitting collateral in either currency contains an option on the basis, which has value.

The resettable notional

One feature of the cross-currency market deserves naming because it solves a problem the rest of this chapter would otherwise leave open. A fixed-notional cross-currency swap exchanges principal at inception and at maturity at the same rate. If the exchange rate moves, the present value of the final exchange moves one-for-one with it, so the swap's mark-to-market grows with the move. Variation margin does cover that mark, but a mark that large means large and growing collateral flows, a large exposure over the margin period of risk, and a large SA-CCR add-on, all on a swap whose principal is paid once at the end.

A **mark-to-market** or **resettable cross-currency swap** fixes this by resetting one leg's notional at each payment date to the current spot equivalent, with the difference paid in cash. The mark is then reset to zero periodically rather than accumulating, and the swap's exposure profile of Section 9.1 is dramatically flatter. Two consequences follow.

The resettable version prices differently, and the difference is a convexity adjustment. Resetting the notional means the interest paid on it depends on the exchange rate at the reset date, so the payment is a product of a rate and an FX level, which is the quanto configuration of Section 30.6. The correction is the covariance between the exchange rate and the interest rate, and it is why resettable and fixed-notional swaps quote at different bases.

The choice is driven by Chapter 27, not by pricing. The resettable structure is now the market standard for interbank cross-currency swaps because it cuts exposure, and with it capital wherever the swap is not margined daily. A convention adopted for capital reasons carried a

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valuation consequence with it, which is the same pattern as Section 33.4's in-arrears adjustment, running in the opposite direction: there a convention change removed a convexity; here one introduced it. Example 33.7 puts both halves of that trade-off in numbers, because the exposure saving is large and obvious while the valuation cost is small and easy to omit.

Example 33.7 What the reset saves and what it costs

A five-year $100 million cross-currency swap paying semiannually, spot 1.10, a 3% euro forward rate with 25% rate volatility and 10% FX volatility. The second table below shows three of the ten payment dates; its average is over all ten.

The saving in margin and exposure is the reason the reset exists. Under a fixed notional, a move in spot changes the swap's mark by the move times the notional. Variation margin covers the mark, but the amount to be margined, and the exposure over the margin period of risk, grow with it. The table shows the size of the mark:

FX move	Mark, fixed notional	Mark after reset
+10%	$10.0m	≈ $0
+20%	$20.0m	≈ $0
+30%	$30.0m	≈ $0

Thirty percent of notional is a very large mark to margin daily, and every day's change in it is exposure over the margin period of risk.

The valuation cost runs the other way and is not zero. Interest accrues on a notional that is itself a function of spot, so each payment is a product of a rate and an FX level and the forward rate must be adjusted by $-F\rho\,\sigma_S\,\sigma_F\,T_i$, where ρ is the correlation between the exchange rate and the euro rate, σ_S and σ_F are their volatilities, and T_i is the reset date, half a year before the payment, that fixes the payment's notional:

Payment date	$\rho = 0$	$\rho = 0.3$	$\rho = 0.6$
1 year	0.00 bp	−1.13 bp	−2.25 bp
3 years	0.00 bp	−5.63 bp	−11.25 bp
5 years	0.00 bp	−10.13 bp	−20.25 bp
Average	**0.00 bp**	**−5.06 bp**	**−10.13 bp**

At a correlation of 0.3 the reset feature is worth about 5 basis points on the floating spread, the same order as the year-on-year inflation adjustment of Section 33.10 and two orders of magnitude larger than the post-transition in-arrears correction. It grows linearly in maturity and linearly in the correlation, and it vanishes only if rates and the exchange rate are independent, which for a currency pair whose basis is a funding variable is not a natural assumption.

So the two effects are not comparable in kind and both are real: tens of millions of mark to margin against single-digit basis points of value. The market chose the exposure saving, correctly, and then had to learn to price the residue.

33.10 INFLATION SWAPS

Inflation swaps are the largest swap market this book has not yet priced, and one where the convexity structure is unusually clean.

A **zero-coupon inflation swap** exchanges a fixed compounded rate for the realized index ratio $I_T/I_0 - 1$, settled once at T. There is one payment and one observation ratio, so there is no timing mismatch and no convexity adjustment. The quoted rate is the **breakeven inflation rate** directly: the fixed inflation rate at which the swap is worth zero, which is the market's price for inflation over the term.

A **year-on-year inflation swap** pays $I_k/I_{k-1} - 1$ each year. Each payment is a ratio of index levels observed at two different dates and paid at the later one, so the earlier level is paid

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a year after its natural date, and that is exactly the configuration Chapter 30 says produces an adjustment. Its size is set by the covariance between the index and nominal rates: with deterministic rates it would vanish, however volatile the index. Example 33.8 sizes it.

Example 33.8 Why the two inflation markets quote separately

Breakeven inflation flat at 2.50%; index volatility 1.6%, normal volatility of the one-year nominal rate 1.0 percentage point, and a correlation of +0.9 between them, since inflation and nominal rates tend to rise together. The adjustment is then $0.9 \times 1.6\% \times 1.0\% = 1.44$ basis points for each year to the earlier fixing, and it is positive.

Year	Year-on-year convexity adjustment
1	0.00 bp
2	1.44 bp
5	5.76 bp
10	12.96 bp

Averaged over ten years, the adjustment is 6.48 basis points.

So a ten-year year-on-year swap does not strike at the ten-year zero-coupon breakeven, and the gap is not a quoting error. It is the accumulated convexity of ten ratio payments. This is why the two inflation markets are quoted independently rather than one being derived from the other. Two features of inflation swaps have no analog in rate swaps.

Indexation lag. The index used for a payment is typically the one published two or three months earlier, because the statistics office needs time. The swap therefore references a rate that is already partly known, which reduces its volatility near each reset and complicates the interpolation between monthly index points.

Seasonality. Consumer price indices have a pronounced and stable annual pattern. A year-on-year swap spanning exactly twelve months is nearly seasonality-neutral; anything else is not, and a curve built without a seasonal adjustment will misprice every off-anniversary payment in a predictable direction.

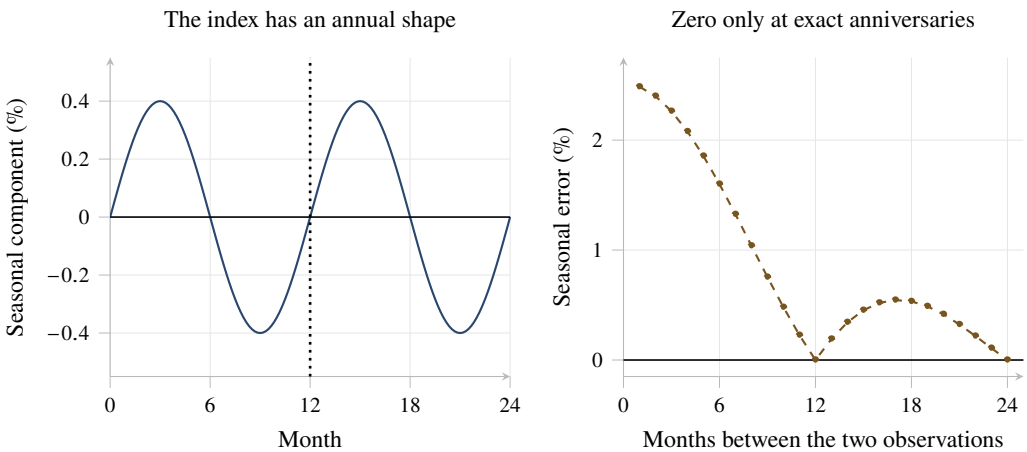


Figure 33.2: Illustrative seasonal amplitude of 0.4%, not market data. Right panel: worst case over the ending month, annualized.

Figure 33.2 is why an inflation curve is quoted by calendar month rather than by tenor. The left panel's pattern cancels exactly when the two observations are twelve months apart, which is what makes a standard year-on-year swap clean. The right panel is what happens when they are not: a swap observing three months apart carries a seasonal error of over two percentage

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points, annualized, and it is a *predictable* error rather than a risk: the same months produce the same sign every year. An unadjusted curve therefore does not misprice off-anniversary payments at random; it misprices them in a direction a counterparty can look up.

The users are worth naming, because the flow is one-directional and it shows in the price. Pension funds and insurers with inflation-linked liabilities are structural receivers of inflation, and there is no natural payer of comparable size; inflation-linked bond issuance by governments is the main one, and it is smaller. So breakeven inflation from swaps has tended to sit above breakeven from bonds (the gap between a nominal government bond's yield and an inflation-linked bond's real yield) in markets such as the sterling market, where pension demand is concentrated, and the gap is a demand imbalance rather than a forecast disagreement. Section 7.9's pension demand compressing swap spreads has the same structure. In sterling, the Retail Prices Index (RPI) is also to be aligned with CPIH, the consumer prices index including owner occupiers' housing costs, in February 2030, as the government and the UK Statistics Authority announced in November 2020; the change reprices long-dated RPI swaps and index-linked bonds alike, and it falls on the pension schemes that hold them.

33.11 EQUITY AND TOTAL RETURN SWAPS

Section 7.12 introduced these and Section 27.14 showed what they made possible. The pricing is the easy part and the funding is not.

An equity swap exchanges the total return on an index or single stock for a floating rate plus a spread. Its fair spread is not a forecast of the equity's return (the equity leg's expected return under the risk-neutral measure is the financing rate, by Chapter 13) but a function of the dealer's cost of holding the hedge:

$$\text{spread} \approx \text{funding} + \text{borrow cost} + \text{dividend leakage} + \text{balance sheet charge}$$

The borrow cost is the fee for borrowing stock to hedge a client's short; **dividend leakage** is the part of the dividends the dealer loses to withholding tax and so cannot pass on; the balance sheet charge is the cost of the capital the hedge position consumes. Not every term applies to every trade. A dealer hedging a client's long buys the stock, so it pays funding and loses the leakage but borrows nothing; one hedging a client's short borrows the stock and sells it, so it receives cash rather than needing it and holds no dividends to lose, and only the borrow cost and the balance sheet charge remain. Each term is observable to the dealer and none is to the client, which is why equity swap spreads vary far more across dealers than swap rates do, and Example 33.9 shows how little a client learns from the quote.

Example 33.9 Four costs behind one number

A one-year total return swap on a $50 million basket yielding 2.0%. The dealer's costs:

Component	bp	Determined by
Funding above the risk-free rate	25	the dealer's own credit
Dividend leakage: 15% withholding on 2.0%	30	the dealer's tax residence
Balance sheet under the leverage ratio	15	Section 27.10's constraint
Quoted spread, synthetic long	**70**	
Stock borrow, if the client is short	40	securities lending supply
Balance sheet under the leverage ratio	15	Section 27.10's constraint
Quoted spread, synthetic short	**55**	

The short-side spread comes off the floating rate the client receives, and it is not the long-side spread plus the borrow: of the long side's three costs only the balance sheet charge survives.

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The client sees one number and cannot decompose it. The 25 bp between dealers quoting 70 and 95 basis points can be assembled in more than one way (the cheaper dealer funding 10 bp below the other and a tax treaty sparing it 15 bp of dividend leakage, say, or the treaty plus a leverage ratio that binds it less), and the client cannot tell which, because none of the three is disclosed and all three move independently of the equity market.

The practical consequences run in both directions. The dividend term is a tax fact presented as a market price, so a client that could hold the shares directly and reclaim withholding is paying 30 bp for a service it does not need. The balance sheet term is the one that moves without warning: it is a function of the dealer's leverage ratio headroom rather than of anything about the trade, so a spread can widen at a quarter-end for reasons the client will never observe. That is Section 27.10's constraint reaching a customer who has never heard of it.

Three uses account for most of the market, and they explain the spread's components.

Synthetic long exposure with leverage, which is the Archegos configuration of Section 27.14. Here the balance-sheet charge is the component that grows with the client's book: on a single trade it is the smallest of the three, as in Example 33.9, but once one client's exposure is large enough to move the dealer's leverage ratio it becomes the cost that limits how much the dealer will extend.

Synthetic short exposure, where the borrow cost dominates and Section 5.2's hard-to-borrow stocks carry spreads of hundreds of basis points.

And dividend and tax positioning, where the swap's treatment of dividends differs from direct ownership. This is the least discussed and historically the most controversial use, and the regulatory attention it has received in several jurisdictions is the reason many dealers now restrict it.

33.12 UNWINDS, NOVATIONS, AND COMPRESSION

A swap can be ended in three ways, and they are not equivalent.

Unwind: the two parties agree on a termination payment equal to the mark-to-market. This is simple, but it requires the original counterparty's cooperation and pricing.

Novation: a third party steps into one side. This preserves the trade and changes the counterparty, and it requires consent because the remaining party is acquiring a new credit exposure, which is Chapter 27's subject appearing as a documentation clause.

Offsetting: enter an opposite trade. The market risk is flat and the counterparty risk is doubled, which is exactly the wrong trade under Chapter 27's framework and is done anyway when an unwind cannot be agreed.

Compression exists because of the third. **Multilateral compression** services tear up offsetting trades across many participants while preserving each one's net market risk. Section 25.18 recorded compression, alongside clearing and a genuine decline in activity, as one of the three forces that cut the CDS market's gross notional by roughly an order of magnitude; in rates it has not shrunk the market but has kept gross notional far below what new trading alone would have produced, so a change in the notional statistics that headline the derivatives market can be a story about compression rather than about activity.

A swap's life does not end when its risk does, and the gap between the two is a capital cost under Chapter 27 and an operational cost always.

33.13 WHO USES NON-STANDARD SWAPS

Corporate borrowers take amortizing and accreting swaps because their debt amortizes: the non-standard swap exists to make the hedge match the exposure, and a bullet swap against an amortizing loan leaves a residual that grows over time.

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Insurers and pension funds are the structural receivers in inflation swaps and the largest users of long-dated forward-starting swaps, for the reason Section 7.9 gave: their liabilities are long, and bond markets are not deep enough to supply the duration they must hedge.

Structured note investors are the ultimate holders of the CMS convexity of Example 33.3, usually without seeing it separated from the coupon.

Hedge funds use equity and total return swaps for leverage, shorting, and, as Section 27.14 documented, for the position invisibility that the format provides.

And dealers sit in the middle of all of it, running the vega that CMS creates, the basis that cross-currency creates, and the balance sheet that equity swaps consume. The recurring pattern of this chapter is that a non-standard swap moves a risk from a client who does not want it to a dealer who prices it, and the price, often a convexity adjustment, is one the client usually cannot compute.

SUMMARY

A non-standard swap that needs more than discounting is usually a standard swap plus a convexity adjustment, and the adjustment is computable, has a reasoned sign, and is often large enough to matter. It arises wherever a payoff is nonlinear in the rate it depends on, or a rate is paid at a time other than its natural one. An amortizing swap priced at its weighted average life shows how a summary of the schedule fails: the par rate weights each maturity by its annuity, so the shortcut's error follows the curve's slope.

The classic LIBOR-in-arrears adjustment grows with maturity, while a compounded-in-arrears SOFR rate is determined and paid at the end of its period, so the adjustment all but vanishes: a convexity adjustment is a property of a convention, not of a market. The complexity moved rather than vanished, since Section 29.4's backward-looking caplet needs an adjustment its forward-looking predecessor did not. On a CMS swap the adjustment is many times the bid–ask spread, and because it depends on a volatility, a linear-looking product carries vega hedged off the whole smile.

Other products carry options inside them. A range accrual coupon is a strip of digitals whose enhancement is short vega, and it fails in the states where its holder most needs income. A cancellable swap is a vanilla plus a Bermudan, and Example 31.9's calibration ambiguity, quoted as a rate, makes model risk a pricing problem rather than only a risk-management one. A commodity swap settles against an average, which removes much of an option's premium and is neutral only inside a linear payoff.

Basis and inflation swaps show what a spread prices. Tenor basis priced bank credit before 2022 and prices much less now, while the cross-currency basis prices funding scarcity and did not shrink. Zero-coupon and year-on-year inflation swaps differ by a convexity adjustment, which is why the two markets quote independently, and pension and insurance demand can put swap breakevens above bond breakevens. An equity swap's spread is built from funding, dividends, borrow and balance sheet, every term observable to the dealer and none to the client.

Unwind, novation, and offsetting are not equivalent: offsetting flattens market risk and doubles counterparty risk, which is why compression exists and why a change in headline notional can describe compression rather than activity. The recurring pattern is a non-standard swap moving a risk from a client who does not want it to a dealer who prices it, at a price, often a convexity adjustment, the client usually cannot compute.

FURTHER READING

The ISDA definitions are the primary source for every convention in this chapter; the compounding provisions of Section 33.3 and the payment-delay conventions of Section 33.4 in particular repay direct reading, because the summaries in textbooks (including this one) elide

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distinctions that matter operationally. Hull's "Swaps Revisited" chapter is a short treatment of the non-standard varieties. On CMS, Hagan's "Convexity Conundrums" is the practitioner standard and develops Example 33.3's replication argument in full. On inflation, Deacon, Derry, and Mirfendereski is the standard book and covers the seasonality and indexation-lag questions of Section 33.10 in the detail they require. On compression and the notional statistics of Section 33.12, the BIS semiannual survey's methodology notes explain what the headline notional counts. For the post-transition conventions, the market-practice recommendations published by the currency-specific working groups are the record of what was adopted, and they differ across currencies in ways the academic literature tends to ignore.

PRACTICE QUESTIONS

33.1 Using Example 33.1, state the amortizing par rate and the error from the weighted-average-life shortcut, and explain why the error is driven by the curve's slope and which weighting of the schedule removes most of it.

33.2 Explain why a forward-starting swap amplifies curve errors, and why compounding at a spread produces an adjustment while flat compounding does not.

33.3 Using Example 33.2, state the classic in-arrears adjustment at ten years and the post-transition adjustment at ten years, and explain why the ratio between them says nothing about rate volatility.

33.4 Using Example 33.3, state the fair fixed rate with and without convexity adjustments and the difference, and explain why a product with a linear-looking payoff carries a vega.

33.5 Explain why a CMS convexity adjustment always has the same sign, and what that implies for the investor in a CMS-linked note.

33.6 Using Example 33.4, state the accrual fraction, the enhanced coupon, and the coupon's value if realized volatility is 50%, then explain what the enhancement actually is.

33.7 Explain why a range accrual's coupon fails in the states where its holder most needs income, and name the earlier section whose structure this repeats.

33.8 Using Example 33.5, state the ambiguity in the quoted cancellable swap rate implied by Example 31.9 and compare it with a typical bid–ask, then explain why this makes model risk a pricing problem.

33.9 Derive the variance ratio for an average of n observations, evaluate it at $n = 21$, and state its limit. Explain why averaging changes an option's value and not the swap's.

33.10 Explain why an implied volatility quoted on an average is not comparable with one quoted on a single fixing.

33.11 Explain why the mark-to-market of a fixed-notional cross-currency swap grows with the exchange rate, what that does to collateral flows and exposure, and what the resettable structure does about it.

33.12 Explain why a resettable cross-currency swap prices differently from a fixed-notional one, and identify the configuration from Chapter 30 that produces the correction.

33.13 Using Example 33.8, state the year-on-year adjustment at ten years and its average, and explain why the zero-coupon market needs no adjustment.

33.14 Explain why, where pension demand for inflation protection is concentrated, inflation swap breakevens can sit above bond breakevens, and why this is a demand imbalance rather than a forecast disagreement.

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33.15 Distinguish unwind, novation, and offsetting, state what each does to market risk and to counterparty risk, and explain why a change in headline notional statistics can reflect compression rather than activity.

FURTHER QUESTIONS

33.16 Example 33.1 finds an 18.1 basis point error from the average-life shortcut. Derive the error's dependence on the curve's shape, determine the curve under which the shortcut is exact, and state how large the error can become on a realistically steep curve.

33.17 Example 33.2 shows the in-arrears adjustment falling by a factor of about sixty. Derive both adjustments from the change-of-numeraire argument, verify the factor, and determine the payment lag at which the post-transition adjustment would again become material.

33.18 Example 33.3 finds 19.13 basis points on a ten-year CMS swap. Determine how this scales with the CMS tenor, the swap's maturity, and the volatility level; then identify the configuration in which the adjustment exceeds 100 basis points and judge whether such a product would be sellable.

33.19 Section 33.5 argues that a CMS-linked note investor is being sold convexity without seeing its price. Design a disclosure that would make the convexity component visible, state what the issuer would have to reveal, and discuss why such disclosure is not standard.

33.20 Example 33.8 attributes the zero-coupon/year-on-year gap to convexity. Construct the arbitrage that would exist if the two markets quoted at the same breakeven, determine what would have to be traded to capture it, and explain what practical frictions prevent the two from being kept in line.

33.21 Section 33.10 says pension demand makes the inflation swap market one-directional. Model the resulting price impact, compare it with Section 7.9's swap-spread compression and Section 26.3's mortgage convexity flows, and state the general conditions under which a structural demand imbalance persists rather than attracting offsetting supply.

33.22 Example 33.4 shows the range accrual coupon falling from 4.95% to 1.81% as volatility rises from 15% to 50%. Derive the coupon's vega, determine the volatility at which it is largest, and explain what in the band's width sets it.

33.23 Example 33.5 converts a 13.1% model ambiguity into 4.2 basis points on a quoted rate. Work out what a dealer should do about an ambiguity several times its bid–ask: consider reserving, widening, hedging the parameter, and declining the trade, and state the conditions under which each is the right response.

33.24 Section 33.9 says the resettable cross-currency structure was adopted for capital reasons and carried a pricing consequence. Derive the covariance correction between the resettable and fixed-notional versions, estimate its size for a major currency pair, and judge whether it is large enough that the capital saving might be partly illusory.

33.25 This chapter argues that a non-standard swap moves risk from a client who cannot price it to a dealer who can. Assess this as a description of the market: identify where the client is compensated for the information asymmetry, where they are not, and what a regulator concerned with the second could usefully require.

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